

Assignment 2 - Part 3: Synchronization of threads

Fall 2026

Due: Sunday, Oct 4, 11:59 pm

Download the starter code A2_3 from D2L and complete the following tasks.

Coding Task 1: Choose a synchronization technique (mutex, condition variables, barrier, or any combination) to correct the problem of the “Neuromancer Cyber Heist Game”. Use neuromancer.c as your starter code.

Hint 1: The currentPlayer tracker is a global variable, and the players are not waiting for their turns to attempt a hack.

Hint 2: You’ll need to use two synchronization techniques to make this a fair game.

Coding Task 2: Choose a synchronization technique (mutex, condition variables, or barrier) to correct the problem of finding the average of an array. Use average.c as your starter code.

Hint: This is a phased computation. The average is being calculated before the summing process is fully complete. Ensure that all threads finish summing before proceeding with the average calculation.

NOTE: Ensure proper creation, initialization, management, and destruction of the chosen synchronization technique.

[Discussion Post] Task 3:

Examine paradox.c and compare run_a() with run_b(). What issues or differences do you notice?

Compile the program using:

```
gcc -O2 paradox.c -lpthread -o paradox
```

Then run the program `./paradox` with 1, 2, 4, 8, and 12 threads. Share the timings and sums. Explain any performance or numerical differences you observe.

Answer the following:

- What were your expectations and what were your observations. How did execution time and final sums change across the tests.
- What in the code or design could explain these results.
- Research ways to redesign and test the program to improve it.

Deliverables:

- A zipped folder named `A2_part_3` submitted to D2L dropbox that contains:
 - o Your source code files for both coding tasks 1 and 2.
 - o A pdf file for screenshots showing your codes fixed the issue and work as expected.
 - o A pdf for Task 3 and 4.
 - o A README.txt file
- Available on GitHub inside the folder where you added me as a collaborator.

Total Points (50)

- Code runs and works as expected – Task 1: 15 points
- Code runs and works as expected – Task 2: 10 points
- Clear, detailed explanation – Task 3: 15 points
- Proper commenting and screenshot of code output: 5 points
- Available on GitHub: 5 points